

AdaptiveFly: Achieving Transparent Control via Bio-Congruent Adaptation for Embodied 3D Locomotion

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概要. Traditional 3D locomotion in Virtual Reality (VR) often fails, as abstract mappings (e.g., gamepads) increase cognitive load, while static embodied interfaces create conflicts by ignoring both inter-person variability in body sway and the context-dependent nature of human sensorimotor coupling. We introduce AdaptiveFly, a 4-DoF embodied flight paradigm utilizing Bio-Congruent Adaptive Mechanisms. AdaptiveFly statistically filters unintentional postural sway to ensure stable, responsive control tailored to each user and provides a speed-dependent adaptive yaw coupling that dynamically resolves the “Look vs. Steer” conflict. Preliminary observations from a public demonstration suggest this dual-adaptive approach yields high intuitiveness and strong signs of embodiment, such as reflex preservation.

1 Introduction

Enabling a rich Sense of Embodiment (SoE) is crucial for interaction in VR[1]. Achieving natural 3D locomotion in XR requires realizing Transparent Control—where the technology disappears, and the user’s intentions are translated into action without conscious effort directed at the interface.

Traditional methods fall short. Gamepads demand high cognitive load and diminish the SoE[2]. Embodied interfaces like leaning-based controls typically utilize Static Mapping[3]. However, they ignore the vast inter-person variability in natural body sway, making them feel unresponsive or overly sensitive. Second, they ignore that human sensorimotor coupling is highly Context-Dependent[4]. This creates the “Look vs. Steer” conflict: when stationary, humans decouple head rotation from direction; when moving quickly, these become tightly coupled. Static systems cannot accommodate both, breaking transparency and SoE[5].

We introduce AdaptiveFly (Fig. 1), a 4-DoF embodied flight paradigm designed to achieve transparent control by mirroring this human phys-

iology. Our system contributes two distinct Bio-Congruent Adaptive Mechanisms to solve both failures of static mapping: 1. Personalized Translation Control: A statistical model that calibrates to each user’s unique, natural sway distribution (Fig. 2), robustly filtering unintentional movement. 2. Context-Adaptive Yaw: A dynamic control manifold (Fig. 3) that tightens head-body coupling as speed increases, resolving the “Look vs. Steer” conflict..

2 The AdaptiveFly System

AdaptiveFly maximizes transparency by ensuring stability and responsiveness across different movement dynamics.

2.1 Personalized Translation Control

We use a force platform to measure the user’s Center of Pressure (CoP) for horizontal translation. To ensure stability, we implement personalized calibration, estimating the user’s natural sway distribution at rest. As shown in Fig. 2, this creates a statistical “smart deadzone.” Translational force (red curve) is applied only when the CoP shift (X-axis) significantly exceeds this personalized sway baseline (blue curve), effectively filtering unintentional sway while remaining responsive to intentional leaning.

2.2 Context-Adaptive Yaw

Our second mechanism addresses the “Look vs. Steer” conflict. Human sensorimotor cou-

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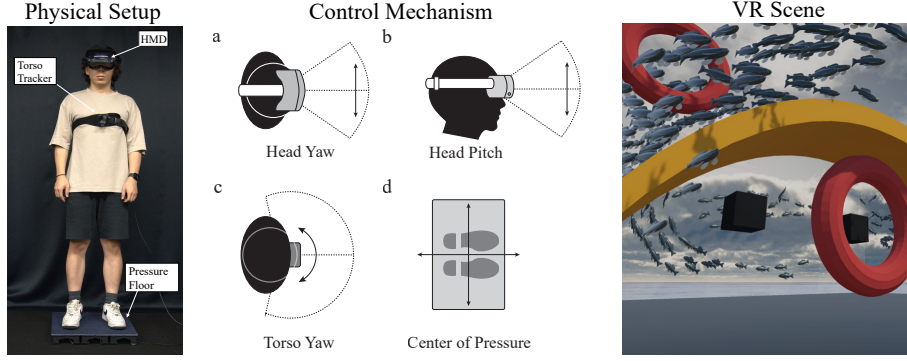


Fig. 1. System Overview

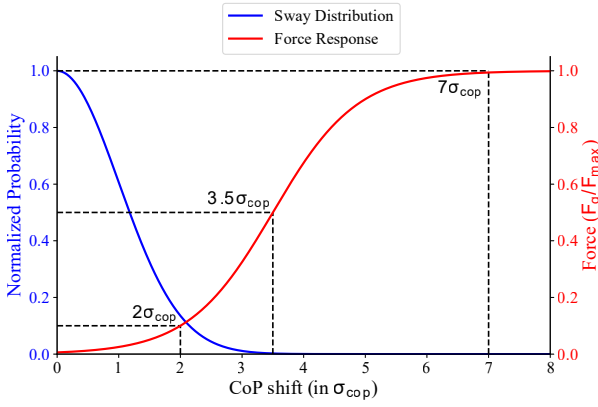


Fig. 2. Response in Force vs. Sway Distribution

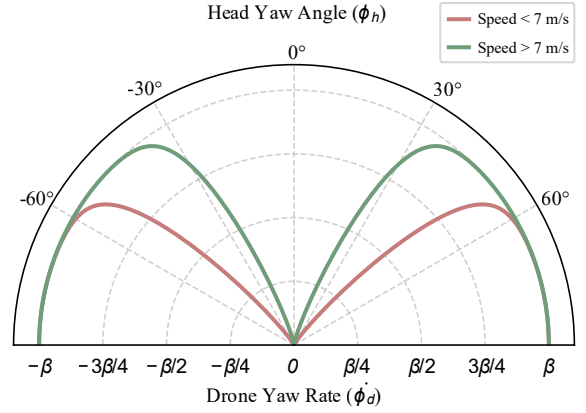


Fig. 3. Adaptive Yaw Control Manifold

pling is context-dependent: at rest, the head moves freely, but during locomotion, a change in direction is tightly coupled with a preceding head rotation[4, 6]. AdaptiveFly mirrors this bio-congruence by using a dynamic control manifold that adapts its “deadzone” size based on current velocity. As shown in Fig. 3, this transparently supports both modes:

Low-speed exploration: A large deadzone (about 30°) allows users to naturally look around without unintended steering.

High-speed travel: A small deadzone about 12° creates tight coupling for precise, responsive steering.

2.3 Asymmetric Pitch Control

Elevation is controlled via head pitch. To improve comfort during vertical maneuvers, we employ an asymmetric mapping. This design is motivated by the finding that human head movements are naturally asymmetrical, with the influence of head movements being smaller for down-

wards gaze shifts compared to those targeted upwards [7, 5]. By adopting this natural asymmetry, our proposed system effectively mitigates unnatural or extreme movements.

3 Discussion and Conclusion

Preliminary observations ($N=40$) from a public demonstration strongly support the AdaptiveFly paradigm. The rapid adoption by naive users highlights the potential for bio-congruent control to reduce training. Furthermore, we observed strong signs of embodiment, such as reflex preservation, where users would instinctively lean away from virtual obstacles. This suggests a successful shift toward Transparent Control, where the interface becomes an extension of the body.

We conclude that AdaptiveFly offers a promising approach to intuitive and embodied 3D navigation. Future work involves formal user studies to quantify its benefits against static mappings and gamepad baselines.

参考文献

- [1] Konstantina Kilteni, Raphaella Groten, and Mel Slater. The sense of embodiment in virtual reality. *Presence*, Vol. 21, No. 4, pp. 373–387, 2012.
- [2] Ashu Adhikari, Abraham M Hashemian, Thinh Nguyen-Vo, Ernst Kruijff, Markus von der Heyde, and Bernhard E Riecke. Lean to fly: leaning-based embodied flying can improve performance and user experience in 3d navigation. *Frontiers in Virtual Reality*, Vol. 2, p. 730334, 2021.
- [3] Abraham M. Hashemian, Matin Lotfaliei, Ashu Adhikari, Ernst Kruijff, and Bernhard E. Riecke. Headjoystick: Improving flying in vr using a novel leaning-based interface. *IEEE Transactions on Visualization and Computer Graphics*, Vol. 28, No. 4, pp. 1792–1809, 2022.
- [4] Takao Imai, Steven T Moore, Theodore Raphan, and Bernard Cohen. Interaction of the body, head, and eyes during walking and turning. *Experimental brain research*, Vol. 136, No. 1, pp. 1–18, 2001.
- [5] Ludwig Sidenmark and Hans Gellersen. Eye, head and torso coordination during gaze shifts in virtual reality. *ACM Trans. Comput.-Hum. Interact.*, Vol. 27, No. 1, December 2019.
- [6] Halim Hicheur, Stéphane Vieilledent, and Alain Berthoz. Head motion in humans alternating between straight and curved walking path: combination of stabilizing and anticipatory orienting mechanisms. *Neuroscience letters*, Vol. 383, No. 1–2, pp. 87–92, 2005.
- [7] K Han Kim, Matthew P Reed, and Bernard J Martin. A model of head movement contribution for gaze transitions. *Ergonomics*, Vol. 53, No. 4, pp. 447–457, 2010.

未来ビジョン

Our future vision is the complete realization of Transparent Control, where the distinction between the user’s physical body and their virtual actions is fully erased. The work presented in AdaptiveFly—using bio-congruent mechanisms for personalized sway and context-adaptive yaw—is merely the first step.

We envision future systems that move beyond 4-DoF to encompass full-body, 6-DoF interaction, modeled entirely on a deep, continuous understanding of the user’s sensorimotor profile. This interface will not be “operated”; it will be inhabited.

Imagine an adaptive system that learns not only the user’s static sway profile but

also their dynamic gait, their anticipatory balance shifts before a jump, and the subtle gaze-head-torso couplings that signal intent to interact, not just to move. By creating systems that are deeply congruent with this full spectrum of human physiology, we can finally break the barrier of conscious operation. Users will no longer “steer” or “control” a virtual body; they will simply act, with their intentions translated seamlessly into the virtual world, achieving a persistent and profound Sense of Embodiment across any task.